

1 The preceding examples involved a single, observable random variable. In
 2 most cases, we have a sequence of random variables in the model.

3 A common case is where $X_1, X_2, \dots, X_n$ are i.i.d. with density $f_X(x; \theta)$,
 4 where θ is the parameter to be estimated. In this case, the joint distribution
 5 of the random variables is

$$6 \quad \prod_{i=1}^n f_{X_i}(x_i; \theta) = f_{(X_1, X_2, \dots, X_n)}(x_1, x_2, \dots, x_n)$$

7 and the likelihood function is

$$8 \quad L(\theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta)$$

1 Finding the MLE Analytically

2 In some situations the MLE can be derived analytically, usually via calcu-
 3 lus, but sometimes via direct inspection of the likelihood.

4 One “trick” that is often very helpful: The likelihood function is positive
 5 at its maximum. Hence, the likelihood is maximized at the same θ as does
 6 the logarithm of the likelihood.

7 Hence, we often work with the **log likelihood** instead of the likelihood.

8 This is often denoted $\ell(\theta) = \log L(\theta)$.

9 In the case where $X_1, X_2, \dots, X_n$ are i.i.d. $f_X(x; \theta)$, then

$$10 \quad \ell(\theta) = \log \underbrace{\prod_{i=1}^n f_{X_i}(x_i; \theta)}_{L(\theta)} = \sum_{i=1}^n \log f_{X_i}(x_i; \theta)$$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Exponential(λ)
 2 distribution. Derive the MLE for λ .

$$f_{X_i}(x_i) = \lambda e^{-\lambda x_i}, \quad x_i > 0$$

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i}, \quad x_i > 0$$

$$= \lambda^n e^{-\lambda \sum x_i}$$

$$\ell(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum x_i$$

$$\frac{\partial \ell(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum x_i$$

set equal to 0, solve for λ , find $\hat{\lambda}_{MLE} = \frac{1}{\bar{X}}$
 (check second derivative)

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Beta($\alpha, 1$) distri-
 2 bution. Derive the MLE for α . Compare this with the MOM estimator.

$$f_X(x; \alpha) = \alpha x^{\alpha-1}, \quad 0 < x < 1$$

$$L(\alpha) = \prod_{i=1}^n \alpha x_i^{\alpha-1} = \alpha^n \left(\prod_{i=1}^n x_i \right)^{\alpha-1}$$

$$\ell(\alpha) = n \log \alpha + (\alpha-1) \log \left(\prod_{i=1}^n x_i \right)$$

$$\frac{\partial \ell(\alpha)}{\partial \alpha} = \frac{n}{\alpha} + \log \left(\prod_{i=1}^n x_i \right)$$

$$\hat{\alpha}_{MLE} = - \frac{n}{\log \left(\prod_{i=1}^n x_i \right)} = - \frac{n}{\sum \log(x_i)}$$

$$E(X) = \frac{\alpha}{\alpha+1}, \quad M_1 = \bar{X}, \quad \text{so} \quad \hat{\alpha}_{mom} = \frac{\bar{X}}{1-\bar{X}}$$

- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. with the $\text{Normal}(\mu, \sigma^2)$ dis-
 2 tribution. Derive the MLE for μ and σ^2 .

$$3 \quad f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

$$4 \quad L((\mu, \sigma^2)) = \prod_{i=1}^n f_{X_i}(x_i)$$

$$5 \quad = \left(\frac{1}{2\pi}\right)^{n/2} \left(\frac{1}{\sigma^2}\right)^{n/2} \exp\left(-\frac{\sum_{i=1}^n (X_i - \mu)^2}{2\sigma^2}\right)$$

$$6 \quad \ell((\mu, \sigma^2)) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{\sum (X_i - \mu)^2}{2\sigma^2}$$

$$7 \quad \frac{\partial \ell}{\partial \mu} = \frac{\sum (X_i - \mu)}{\sigma^2}$$

$$1 \quad \text{set equal to 0, find } \hat{\mu}_{MLE} = \bar{X}$$

$$2 \quad \frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum (X_i - \mu)^2}{2\sigma^4}$$

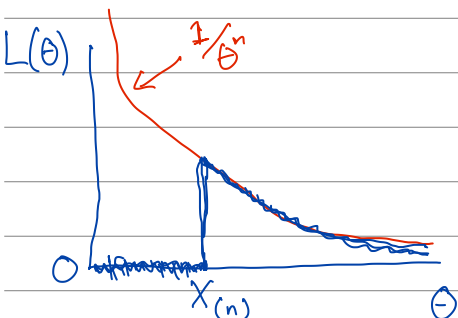
$$3 \quad \text{Find } \hat{\sigma}_{MLE}^2 = \frac{\sum (X_i - \bar{X})^2}{n} \neq \text{sample variance}$$

4
 5 Should check "second deriv." $\rightarrow$ check
 6 that Hessian is negative definite
 7

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Uniform}(0, \theta)$. Derive the
 2 MLE for θ . What is the MOM in this case?

$$f_x(x) = \frac{1}{\theta} \mathbb{1}_{\{0 \leq x \leq \theta\}}$$

$$L(\theta) = \prod_{i=1}^n f_{X_i}(x_i) = \frac{1}{\theta^n} \prod_{i=1}^n \mathbb{1}_{\{0 \leq x_i \leq \theta\}}$$

$$L(\theta) = \begin{cases} \frac{1}{\theta^n}, & 0 \leq x_i \leq \theta \text{ for all } i \\ 0, & \text{otherwise} \end{cases}$$


$$\hat{\theta}_{MLE} = X_{(n)}$$

1 Clearly biased. In fact, $\theta \geq X_{(n)} = \hat{\theta}_{MLE}$

$$E(X_i) = \frac{\theta}{2}, \quad \hat{\theta}_{mom} = 2\bar{X}$$

2 This is unbiased, but $2\bar{X}$ could
 3 be less than $X_{(n)}$

Part 5: Properties of MLEs

Summary

In this Part we explore some of the important properties possessed by maximum likelihood estimators.

Briefly stated, we will see that, asymptotically, these estimators cannot be beat, in the sense that they have minimal MSE, and also are approximately normally distributed with variance that is easy to calculate. This makes quantifying the error in the estimators straightforward.

Invariance of the MLE

The invariance result can be stated as follows: If $\hat{\theta}_{\text{MLE}}$ is the MLE for θ , then $g(\hat{\theta}_{\text{MLE}})$ is the MLE for $g(\theta)$.

This is a useful, important result because it is often the case that we are truly interested in estimating some function of the parameter θ , not θ itself.

Using this, we can say, for example, that if $X_1, X_2, \dots, X_n$ are i.i.d. normal(μ, σ^2), then the MLE for σ is

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

and the MLE for the **signal to noise ratio** μ/σ is

$$\bar{X} / \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- 1 One consequence of this property is that estimation is not affected by how
 2 one chooses to **parameterize** a model.
- 3 For example, some choose to represent the Exponential distribution using
 4 the “mean parameterization” which leads to $E(X) = \mu$, whereas we have
 5 used the “rate parameterization” which leads to $E(X) = 1/\lambda$. Because
 6 of invariance, these two approaches will lead to the same estimates, i.e.,
 7 $\hat{\mu} = 1/\hat{\lambda}$.

- 1 **Exercise:** Does unbiasedness have this invariance property?

2 No. $\hat{\theta}$ unbiased for θ does not imply
 3 $g(\hat{\theta})$ is unbiased for $g(\theta)$, e.g. s is
 4 not unbiased for σ

- 6 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$. What is the
 7 MLE of the **tail probability** $P(X_i > a)$? $a > 0$

8 Could let $\tau = P(X_i > a)$ and rewrite pdf as a
 9 function of τ . Maximize likelihood over τ

10 Or: Could use invariance and say that
 11 $\hat{\tau}_{MLE} = e^{-\hat{\lambda}a} = e^{-a/\bar{x}}$

1 Consistency of the MLE

2 An estimator $\hat{\theta}$ is **consistent for θ** if as the sample size n increases, the esti-
3 mator converges to the true value of the parameter.

4 Formally, we would write: For any $\epsilon > 0$,

$$5 \quad \lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta_0| > \epsilon) = 0$$

6 This is a fairly low standard placed on an estimator, and, fortunately, under
7 certain “regularity conditions,” the MLE is consistent.

1 Asymptotic Normality of the MLE

2 Another great property of the MLE: The MLE is asymptotically normal,
3 with variance that is not difficult to calculate, under certain “regularity
4 conditions.” (See page 516 from Casella and Berger, Second Edition.)

5 This will allow an easy route to:

- 6 1. Calculate standard errors for MLEs
- 7 2. Constructing confidence intervals for parameters
- 8 3. Constructing confidence intervals for general functions of parameters
9 (via the Delta method)
- 10 4. Constructing hypothesis tests for parameters

1 To start, we focus on the case where the parameter θ is one-dimensional.

2 The formal result can be stated as follows:

3 If $X_1, X_2, \dots, X_n$ are i.i.d. $f(x; \theta_0)$, then, under suitable regularity condi-
 4 tions, as n increases,

$$5 \quad \sqrt{n}(\hat{\theta} - \theta_0) \rightarrow N(0, I^{-1}(\theta_0))$$

6 where

$$7 \quad I(\theta_0) = E\left(-\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2}\right)$$

← assumed to $f(x; \theta_0)$

8 is the **Fisher Information**.

1 **Exercise:** Explain the practical value of the above result.

$$2 \quad \sqrt{n}(\hat{\theta} - \theta_0) \rightarrow N(0, I^{-1}(\theta_0))$$

$$3 \quad \sqrt{n}(\hat{\theta} - \theta_0) \approx N(0, I^{-1}(\theta_0)) \quad \text{for } n \text{ "large"}$$

$$5 \quad \hat{\theta} \approx N(\theta_0, \frac{1}{nI(\theta_0)})$$

$$6 \quad \text{So, } E(\hat{\theta}) \approx \theta_0 \quad (\text{approx. unbiased})$$

$$8 \quad V(\hat{\theta}) \approx \frac{1}{nI(\theta_0)} \quad \text{In practice, } V(\hat{\theta}) \approx \frac{1}{nI(\hat{\theta})}$$